Hao (Mark) Chen

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RESEARCH INTERESTS

My research focuses on optimizing ML algorithms by efficient algorithm (e.g., LLM quantization, speculative decoding, model merging, expert skipping) and system (e.g., memory management, request scheduling, heterogeneous computing) designs. Key research contributions are highlighted below:

- **Optimized intelligent LLM deployment on resource-constrained devices** by empowering efficient models with test-time scaling and tackling the resulting system challenges through speculative generation, cache-aware scheduling, and dynamic memory allocation, achieving **2× faster** agentic reasoning without cloud infrastructure. (FastTTS, ASPLOS 2026)
- **Made multi-model deployment practical through scalable model merging** by introducing a Frank-Wolfe optimization method that merges multiple expert models into a single model retaining multi-task knowledge, with **no additional training or inference cost**. (FW-Merging, ICCV 2025)
- **Proved that multi-model collaboration improves LLM efficiency** by introducing a progressive mixed-precision decoding strategy where models of varying precision collaborate during autoregressive generation, reducing inference memory with negligible quality loss. (PMPD, ICLR 2025)

EDUCATION

Imperial College London
PhD in Computing — *Sep 2024 – June 2028 (expected)*
Advisor: Prof. Wayne Luk, Dr. Hongxiang Fan

Imperial College London
MEng in Computing — *Oct 2020 – June 2024*
Advisor: Prof. Wayne Luk, Dr. Hongxiang Fan

PUBLICATIONS & MANUSCRIPTS

Conference Papers

- **Hao Mark Chen**, Zhiwen Mo, Guanxi Lu, Shuang Liang, Lingxiao Ma, Wayne Luk, Hongxiang Fan. FastTTS: Accelerating Test-Time Scaling for Edge LLM Reasoning. Accepted at *2026 International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*. [Paper][Code]
- **Hao Mark Chen**, Guanxi Lu, Yasuyuki Okoshi, Zhiwen Mo, Masato Motomura, Hongxiang Fan. Rethinking Optimal Verification Granularity for Compute-Efficient Test-Time Scaling. Accepted at *2025 Conference on Neural Information Processing Systems (NeurIPS)*. [Paper] [Code]
- **Hao (Mark) Chen**, Wayne Luk, Ka Fai Cedric Yiu, Rui Li, Konstantin Mishchenko, Stylianos I Venieris, Hongxiang Fan. Hardware-Aware Parallel Prompt Decoding for Memory-Efficient Acceleration of LLM Inference. Accepted at *2025 Empirical Methods in Natural Language Processing (EMNLP)*. [Paper] [Code]
- **Hao Mark Chen**, Shell Xu Hu, Wayne Luk, Timothy Hospedales, Hongxiang Fan. FW-Merging: Scaling model merging with frank-wolfe optimization. Accepted at *2025 International Conference on Computer Vision (ICCV)*. [Paper] [Code]
- **Hao Mark Chen**, Fuwen Tan, Alexandros Kouris, Royson Lee, Hongxiang Fan, Stylianos I Venieris. Progressive Mixed-Precision Decoding for Efficient LLM Inference. Accepted at *2025 International Conference on Learning Representations (ICLR)*. [Paper] [Code]
- Guanxi Lu, **Hao Mark Chen**, Yuto Karashima, Zhican Wang, Daichi Fujiki, Hongxiang Fan. AdaBlockdLLM: Semantic-Aware Diffusion LLM Inference via Adaptive Block Size. Accepted at *2026 International Conference on Learning Representations (ICLR)*. [Paper][code]
- Charlie Campbell, **Hao Mark Chen**, Wayne Luk, Hongxiang Fan. Enhancing LLM-based Quantum Code Generation with Multi-Agent Optimization and Quantum Error Correction. Accepted at *2025 Design Automation Conference (DAC)*. [Paper]

- Jiahao Gai, **Hao (Mark) Chen**, Zhican Wang, Hongyu Zhou, Wanru Zhao, Nicholas Lane, Hongxiang Fan. Exploring Code Language Models for Automated HLS-based Hardware Generation: Benchmark, Infrastructure and Analysis. Accepted at *2025 Asia and South Pacific Design Automation Conference (ASP-DAC)*. [Paper]
- Zehuan Zhang, Hongxiang Fan, **Hao Chen**, Lukasz Dudziak, Wayne Luk. Hardware-Aware Neural Dropout Search for Reliable Uncertainty Prediction on FPGA. Accepted at *2024 Design Automation Conference (DAC)*. [Paper]
- **Hao (Mark) Chen**, Taowen Liu, Songyun Hu, Leyang Yu, Yiqi Li, Sihan Tao, Jacqueline Lee, Ahmed E. Fetit. Web-Based AI System for Medical Image Segmentation. Accepted at *2023 Conference on Medical Image Understanding and Analysis (MIUA)*. [Paper]
- Hongxiang Fan, **Hao (Mark) Chen**, Liam Castelli, Zhiqiang Que, He Li, Kenneth Long, Wayne Luk. When Monte-Carlo Dropout Meets Multi-Exit: Optimizing Bayesian Neural Networks on FPGA. Accepted at *2023 Design Automation Conference (DAC)*. [Paper]

Arxiv Papers

- **Hao Mark Chen**, Zhiwen Mo, Royson Lee, Qianzhou Wang, Da Li, Shell Xu Hu, Wayne Luk, Timothy Hospedales, Hongxiang Fan. Dynamic Expert Sharing: Decoupling Memory from Parallelism in Mixture-of-Experts Diffusion LLMs. [Paper]
- **Hao Mark Chen**, Zehuan Zhang, Wanru Zhao, Nicholas Lane, Hongxiang Fan. Advancing AI-assisted Hardware Design with Hierarchical Decentralized Training and Personalized Inference-Time Optimization. [Paper]
- **Hao (Mark) Chen**, Liam Castelli, Martin Ferianc, Shuanglong Liu, Wayne Luk, Hongxiang Fan. Algorithm and Hardware Co-Design for Multi-Exit Dropout-based Bayesian Neural Networks. [Paper]

INDUSTRIAL EXPERIENCE

Samsung AI Center Cambridge, UK
Research Intern *July 2025 – December 2025*
Worked on efficient diffusion language model.

Samsung AI Center Cambridge, UK
Research Intern *July 2024 – December 2024*
Worked on efficient LLM inference involving quantization, leading to a publication at *ICLR 2025*.

Qube RT London, UK
Quantative Technologist Intern *April 2023 – Sep 2023*
Developed C++ performance monitoring system for low-latency trading platform.

Ampere Computing Shanghai, China
Java Software Developer *June 2021 – Sep 2021*
Developed a log search engine for Jenkins based on Lucene Search.

AWARDS

- **Microsoft Research Fellow** 2026
- **President's PhD scholarship** 2024 – 2028
- **Winton Capital Applied Undergraduate Project Computing Prize** 2024
- **Governors' MEng Prize in Computing** (the best overall performance in MEng programme) 2024
- **Dean's List (Year 4)**, Imperial College London 2024
- **Dean's List (Year 3)**, Imperial College London 2023
- **Dean's List (Year 2)**, Imperial College London 2022
- **Dean's List (Year 1)**, Imperial College London 2021
- **G-Research Ltd Prize**, G-Research Ltd, 2021
- **Singapore SM1 Scholarship**, Ministry of Education Singapore 2015 – 2019

LEADERSHIP & SERVICE

- Supervised and mentored MEng/BEng students on research projects at Imperial College London, resulting in publications at top venues including ICLR 2026 and DAC 2025.
- Served as reviewer for leading ML conferences and journals: NeurIPS, ICLR, ICML, and TMLR.
- Teaching Assistant at Imperial College London, leading small-group tutorials on discrete mathematics, logic, and graph algorithms for undergraduate computing students.

SKILLS

- **Programming:** C++, C, Python, Scala, Java, Swift, Haskell, Bash
- **Tools:** GCC, Jenkins, Github, Docker, Heroku, AWS
- **Framework:** PyTorch, Keras, ODB, Kitura, Lucene, Jinja2
- **Language:** GRE - 333/340 + 5/6 (Verbal 163, Quantitative 170, Analytical Writing 5); TOEFL iBT 113/120 (Reading 30, Listening 29, Speaking 26, Writing 28)